

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (C.E.)

May-2018

CE322 Data Warehousing & Data Mining (DW&DM)

Date: 08.05.2018, Tuesday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.

SECTION-I

Q-1 Answer the following questions.

1. Is Normalization as important in designing a Data Warehouse as while designing Database? Is Data Normalization recommended for Data Warehouse? Why? 4
2. Can it be possible to say "Data Mining is a simple transformation of technology from Database, Statistics and Machine Learning"? Can all Data Mining functionalities perform alternatively by Data Query Processing or Statistical Analysis? 4
3. Where do you construct data mart instead of data warehouse? 3

Q-2

- [A] What are the criteria by which OLTP is compared with OLAP? 4
- [B] What are the steps involved in Data Transformation for making data suitable for Data Mining? 4

OR

- [B] Data cleaning routines can be used to fill in missing values, smooth noisy data, identify outliers and correct the data inconsistency. Explain how to deal with each of them in terms of problems, in brief. 4
- [C] Explain the primitives for specifying the data mining tasks. 4

OR

- [C] What is a data cube? The weather bureau has about 10,000 probes which are scattered throughout various land and sea locations across the country to collect data such as air pressure and temperature at each hour. All the data have to be stored at a central office of the bureau. Give a 4-D view clearly mentioning the dimensions of the data collected at the central office. 4

Q-3

- [A] In data warehouse technology, a multiple dimensional view can be implemented by a relational database technique (ROLAP), or by a multidimensional database technique (MOLAP), or by a hybrid database technique (HOLAP), Briefly describe each implementation technique. 4

OR

- [A] Why Entity-Relation data model does not suits for data warehouse? Which schema models are used for the data warehouse? 4
- [B] Describe Three-Tier data warehouse architecture. What are cuboids? Use example to illustrate its use in data warehouse implementation. 4

OR

- [B] Which are the pattern interestingness criteria. Define support and confidence. For a company-X, min_sup is 10% and min_conf is 75%, while for another company-Y, min_sup is 20% and min_conf is 60%. For which of the two companies, the rule $A \rightarrow B$ is more strong? Why? 4
- [C] Suppose there is a purchase data of games and videos. Out of 10,000 transactions analyzed, 6000 customer transactions included game, 7500 included videos and 4000 included both game and videos. Prepare a cross (contingency) table and check which type of correlation exists between game and video. 4

OR

- [C] Explain in brief the categorization of major Clustering Methods. List the various Clustering Methods in each category. Give example in each case. 4

SECTION-II

Q-4 Answer the following questions.

1. Efficiency, Accuracy and Scalability are the important challenges to data mining. Briefly discuss. 3
2. What is Neural Networks (NN)? What is Support Vector Machine (SVM)? What are the weaknesses and assumptions if any in both the methods? Compare them also. 4
3. Explain in brief the measure of central tendency. What are the properties of variance and standard deviation and give its formula? 4

Q-5

- [A] What sorts of summary statistics would be of interest to the user? Describe how a box plot can give information about whether the value of an attribute is symmetrically distributed. 4
- [B] Following is the series of data for the price:
80,40,210,150,210,240,340,280,210,390. 4
1. Partition the data into equi-depth bins
 2. Smooth the data by bin means and find outliers, if any.
- [C] Suppose the data for analysis include the attribute age. The age values for data tuple are: **13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70.** 4
1. What are the mean, median and mode of the data.
 2. What is the midrange of the data
 3. Find first quartile (Q1) and third quartile (Q3) of the data.
 4. Show box-plot of the data

OR

Q-5

- [A] Discuss which method out of Data Cube Based Implementation and AOI is most efficient and under which conditions? 4
- [B] How to estimate the probabilities with Bayes' Theorem? Explain with an example Naïve Bayesian Classification. 4

- [C] Where are supervised and unsupervised learnings used and what purpose do they serve? Is classification a supervised or an unsupervised form of learning? What do you mean by accuracy of a classifier? 4

Q-6

- [A] Compare the advantages and disadvantages of eager classification (e.g., decision tree, Bayesian, neural network) versus lazy classification (e.g., k-nearest neighbor, case-based reasoning). 4

OR

- [A] What is attribute relevance analysis? Where is it used? Describe entropy-based method to compute information-gain. 4
- [B] Generate the frequent-3 item-sets with candidate generation for the following transaction database. 4

T1	I1,I2,I5
T2	I2,I4
T3	I1,I2,I4
T4	I1,I3
T5	I2,I3
T6	I1,I3
T7	I1,I2,I3,I5
T8	I1,I2,I3

OR

- [B] The support and confidence measures are insufficient at filtering out uninteresting association rules. Explain with an example how correlation can be used to overcome weakness of the support and confidence. 4
- [C] Briefly explain 2-steps procedure for classification. Distinguish between Classification and Clustering. What are the similarities and differences between them? Can we replace/use one in place of another? 4

OR

- [C] Which are the requirements of a good clustering algorithm? What are typical applications of Clustering? 4